

Maths Question Waec Candidate/Private

Question 1

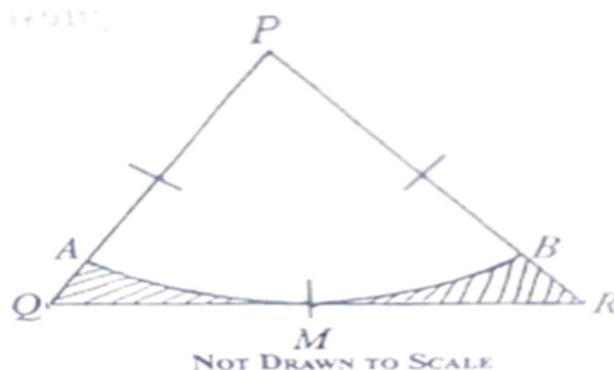
- (a) A car travels a distance of 112 km at an average speed of 70km/h. It then travels further for 60 km at an average speed of 50 km/h. Calculate, for the entire journey, the total time taken.
- (b) If $\frac{x}{y} = 2$ and $\frac{y}{z} = 3$, find the value of $\frac{x+y}{y+z}$.

Question 2

In a football match, the tickets for children and adults were sold at D 3.00 and D 5.00 respectively. If 400 people attended the football match and D 1,700.00 was collected in ticket sales:

1. How many tickets were sold to adults?
2. Mr. Sonko sold 250 tickets. If 175 of the tickets were for adults, how much sales did he make altogether?

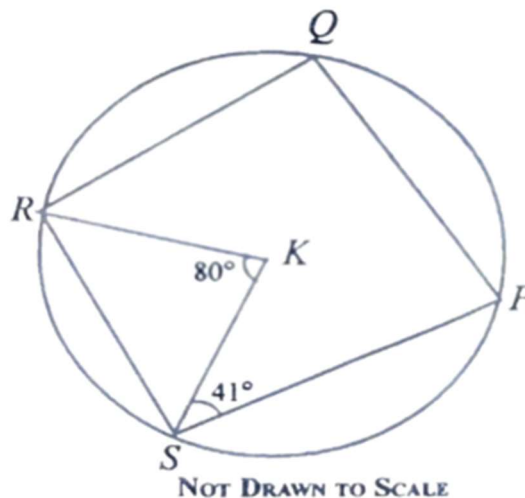
Question 3



In the diagram, PQR is an equilateral triangle of side 18 cm. M is the midpoint of $\overline{QR}$. An arc of a circle with centre P touches $\overline{QR}$ at M and meets $\overline{PQ}$ at A and $\overline{PR}$ at B. Calculate, correct to two decimal places, the area of the shaded region.

[Take $\pi = \frac{22}{7}$]

Question 4



In the diagram, P , Q , R and S are points on the circle centre K . $\overline{KR}$ is a bisector of $\angle SRQ$, $\angle SRQ = 41^\circ$ and $\angle SKR = 80^\circ$. Find:

- (a) $\angle RQP$;
- (b) $\angle SPQ$.

Question 6

A boy stands at a point M on the same horizontal level as the foot, T , of a vertical building. He observes an object on the top, P of the building at an angle of elevation of 66° . He moves directly backwards to a new point C and observes the same object at an angle of elevation of 53° . If $|MT| = 50 \text{ m}$,

- (a) Illustrate the information in a diagram;
- (b) Calculate, correct to **one** decimal place:
 - (i) the height of the building;
 - (ii) $|MC|$.

Question 7

- (a) $M = \{n: 2n - 3 \leq 37\}$, where n is a counting number.
 - (i) Write down all the elements in M .
 - (ii) If a number is selected at random from M , what is the probability that it is a:
 - (α) multiple of 3;
 - (β) factor of 10.
- (b) A shop owner gave an end-of-year bonus to two of his attendants, Kontor and Gapson in the ratio of their ages. Kontor's age is **one and half** times that of Kontor who is 20 years old. If Kontor received Le 200, 000.00, find:
 - (i) the total amount shared;
 - (ii) Gapson's share.

Question 8

1. The sum of three numbers is 81. The second number is **twice** the first. Given that the third number is 6 more than the second, find the numbers.
2. Given the points P(3, 5) and Q on the Cartesian plane such that R is the midpoint of P and Q, find the equation of the line that passes through R and is perpendicular to PQ.

Question 9

(a) Copy and complete the table of values for $y = 2x^2 - x - 4$ for $-3 \leq x \leq 3$.

x	-3	-2	-1	0	1	2	3
y	17			-4			

(b) Using a scale of 2 cm to 1 unit on the x -axis and 2 cm to 2 units on the y -axis, draw the graph of $y = 2x^2 - x - 4$ for $-3 \leq x \leq 3$

(c) Use the graph to find the:

- (i) roots of the equation $2x^2 - x - 4 = 0$;
- (ii) values of x for which y increases as x increases,
- (iii) minimum point of y .

Question 10

The table shows the height of teak trees harvested by a farmer.

Height (m)	3	4	5	6	7	8
Number of trees	4	6	4	5	6	2

1. Find the median height.
2. Calculate, correct to one decimal place, the:
□ . Mean, 2. Standard deviation.

Question 11

In a town, Chief X resides 60 m away on a bearing of 057° from the palace P, while Chief Y resides on a bearing of 150° from the same palace P. The residence of X and Y are 180 m apart.

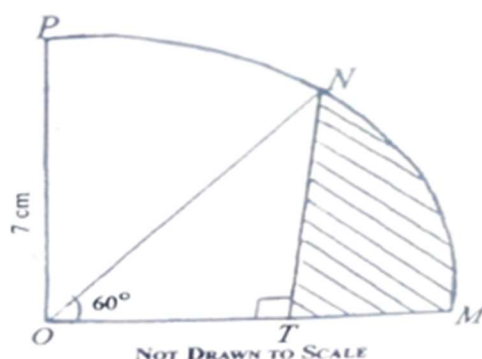
1. Illustrate the information in a diagram.
2. Find, correct to **three** significant figures, the:
□ . Bearing of X and Y;
□ . Distance between P and Y

Question 12

1. Two regular polygons P and Q are such that the number of sides of P is twice the number of sides of Q. The difference between the exterior angle of Q and P is 45° . Find the number of sides of P.
2. The area of a semi-circle is 32. Find, in terms of π , the circumference of the semi-circle.

Question 13

(a)



In the diagram, the radius of the sector of circle centre O, is 7 cm and $\angle MON$ is 60° . Find, correct to one decimal place, the area of the shaded portion.

[Take $\pi = \frac{22}{7}$]

(b) The x and y intercepts of a straight line are $-\frac{3}{4}$ and $\frac{2}{7}$ respectively. Find the equation of the line.

Question 14

A village estimates N1,049,740.00 for a water project. The sum of N400,000.00 is realized through communal donation and deposited in a bank at a compound interest of 10% per annum. If the village chief adds N100,000.00 to the amount in the account at the end of each year, how long will it take to realize the estimate?

Question 14

1. If 3 and $-\frac{2}{7}$ are the roots of the quadratic equation $px^2 + qx + r = 0$, find the values of p , q , and r .
2. One of the parallel sides of a trapezium is one-third the length of the other. If the height of the trapezium is 5 cm and its area is 120 cm^2 , find the lengths of the parallel sides.

Question 15

An isosceles triangle PQR has its vertices on the circumference of a circle.

If $|\overline{PQ}| = |\overline{QR}| = 17 \text{ cm}$, $|\overline{PR}| = 16 \text{ cm}$ and M is the midpoint of $|\overline{PR}|$, calculate:

(a) $|\overline{QM}|$;

(b) correct to the **nearest** whole number, the radius of the circle.

Question 16

(a) The gradient of the line joining points $(-2, n)$ and $(n, 5)$ is $\frac{4}{3}$. Find the value of n .

(b) Given that $\tan x = \frac{15}{8}$, $0^\circ < x < 90^\circ$, find the value of $5\sin x - 6\cos x$.

Question 17

The heights of 10 students in a class are given as follows:

120, 125, 110, 128, 130, 135, 140, 145, 142, and 120.

- (a) Find the range.
- (b) Find the interquartile range.
- (c) Calculate the mean.

Question 18

(a) Aman spent $\frac{1}{3}$ of his income on rent, $\frac{1}{6}$ of the income in a fast food shop and $\frac{3}{4}$ of the remaining amount was saved in a bank. If he had N125, 000.00 left, find :

- (i) his income;
- (ii) the amount saved in the bank.

(b) Solve: $\frac{2+3x}{5} - \frac{1-7x}{3} < 2x + 9$.

Question 19

The heights of students in a class form an Arithmetic Progression (A.P.) such that the tallest student is 1.90 m. The common difference in their heights is 0.05 m and the sum of their heights is 23.25 m.

- (a) How many students are in the class?
- (b) Find the height of the shortest student.

Question 20

1. A tree is 8 km due south of a building. Musa is standing 8 km due west of the tree. (i) Illustrate the information on a diagram. (ii) How far is Musa from the building? (iii) Find the bearing of Musa from the building.
2. If a man rides a bicycle from his house at 5 km/hr, he will get to the office 45 minutes later than when he rides at 6 km/hr. Calculate the distance between his house and the office.

Question 21

(a) A cylindrical tank of diameter 3.5 m is full of water. Three tankers, each of capacity 6,500 litres, draw water from the tank. Calculate, in metres, correct to **two** decimal places, the reduction in height of water in the tank. [Take $\pi = \frac{22}{7}$]

(b) The coordinates of two points P and Q in a plane are (2, -5) and (-4, -6), respectively. Find the equation of the straight line joining the points.

Question 22

(a) A cylindrical tank of diameter 3.5 m is full of water. Three tankers, each of capacity 6,500 litres, draw water from the tank. Calculate, in metres, correct to **two** decimal places, the reduction in height of water in the tank.

$$[\text{Take } \pi = \frac{22}{7}]$$

(b) The coordinates of two points P and Q in a plane are (2, -5) and (-4, -6), respectively. Find the equation of the straight line joining the points.

Question 23

In a large library, Adamu is seated 15 meters away from Uju on a bearing of [specify bearing]. Bola is 25 meters away from Adamu on a bearing of [specify bearing].

1. Illustrate the information in a diagram.
2. Calculate the distance between Bola and Uju.
3. Calculate the bearing of Bola from Uju.

Question 24

The table shows the frequency distribution of the ages of fifty members of a family.

Ages (years)	1-5	6-10	11-15	16-20	21-25	26-30
Frequency						10

Calculate, correct to two decimal places, the:

1. Mean
2. Mean deviation of the distribution

Question 25

Given that $y = \left(\frac{pr}{m} - p^2r\right)^{-\frac{3}{2}}$,

(a) make r the subject;

(b) find the value of r when $y = -8$, $m = 1$ and $p = 3$.

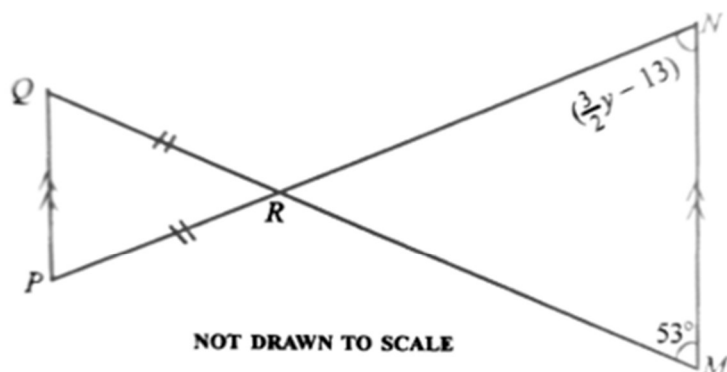
Question 26

A chord subtends an angle of 72° at the centre of a circle of radius 24.5 m. Calculate the perimeter of the minor segment.

$$[\text{Take } \pi = \frac{22}{7}]$$

Question 27

(a)



In the diagram, $\overline{PQ} \parallel \overline{MN}$, $\overline{PR} \parallel \overline{MR}$, $\angle QMN = 53^\circ$ and $\angle MNP = \left(\frac{3}{2}y - 13\right)^\circ$. Find:

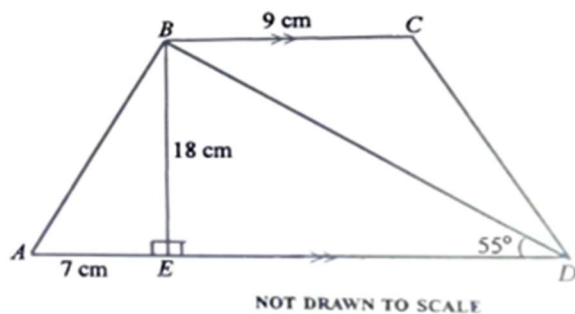
- $\angle PRQ$;
- the value of y .

(b) A bowl contains 15 yellow and green balls of the same size. If the probability of selecting a yellow ball is $\frac{2}{5}$, find the number of:

- yellow balls;
- green balls in the bowl.

Question 28

(a)

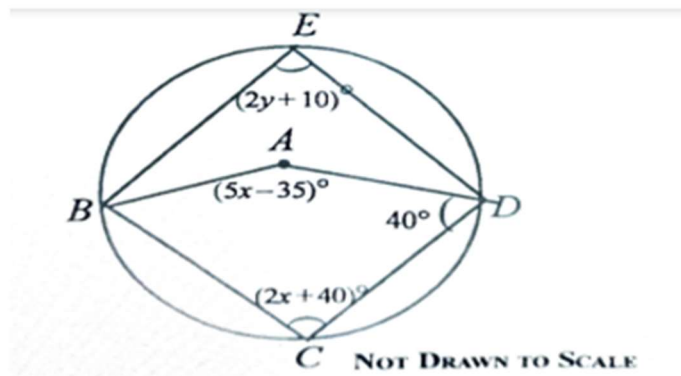


$ABCD$ is a trapezium with $\overline{BC} \parallel \overline{AD}$. E is a point on AD such that $\overline{BE}$ is perpendicular to $\overline{AD}$. If $\angle BDA = 55^\circ$, $|AE| = 7 \text{ cm}$, $|BE| = 18 \text{ cm}$ and $|BC| = 9 \text{ cm}$, find:

- $\angle BAE$;
- the area of $ABCD$.

(b) A trader bought goods for GH¢ x and sold them for GH¢ y . If the profit was 25%, find an equation connecting x and y .

Question 29



In the diagram, $BCDE$ is a circle with centre A . $\angle BCD = (2x + 40)^\circ$, $\angle BAD = (5x - 35)^\circ$, $\angle BED = (2y + 10)^\circ$ and $\angle ADC = 40^\circ$. Find:

- the values of x and y .
- $\angle ABC$.

Question 30

- Given that $m = \tan 30^\circ$ and $n = \tan 45^\circ$, simplify, without using calculator, $\frac{m-n}{mn}$, leaving the answer in the form $p + \sqrt{q}$.
- There are 20 women in a bus. 15 of them wear glasses and 10 wear wrist watches. If a woman is chosen at random from the bus, find the probability that she wears **both** glasses and wrist watch.

Question 31

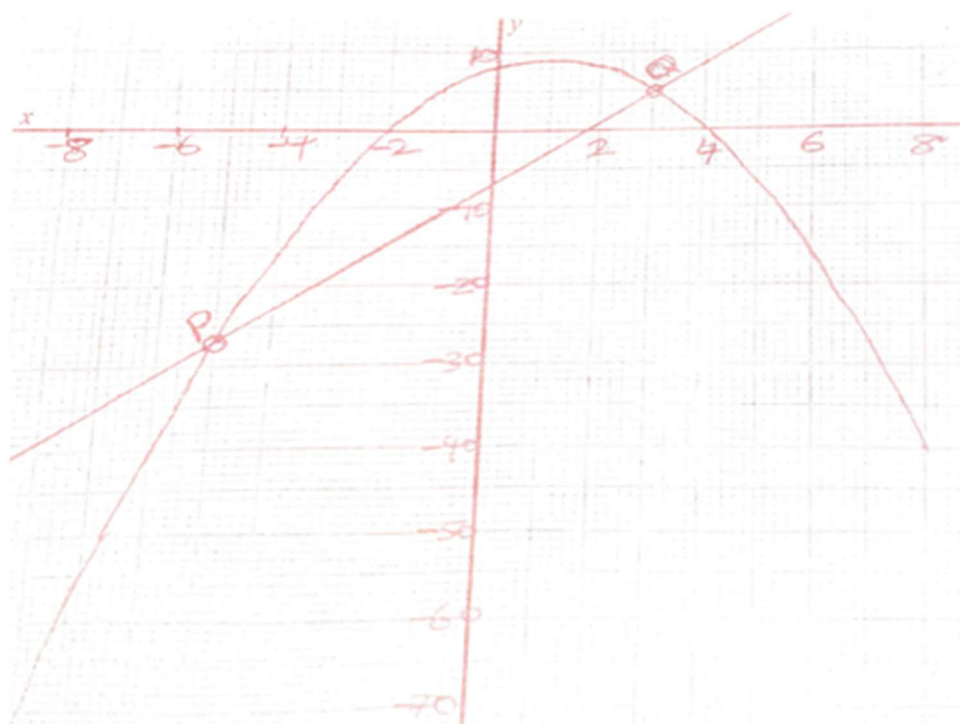
- A man purchased 180 copies of a book at ₦ 250.00 **each**. He sold y copies at ₦ 300.00 **each** and the rest at a discount of 5 kobo in the Naira of the cost price. If he made a profit of ₦ 7,125.00, find the value of y .
- A trader bought x bags of rice at a cost, $c = 24x + 103$ and sold them at a price, $s = 33x - \frac{x^2}{20}$.
 - Find the expression for the profit.
 - If 20 bags of rice were sold, calculate the percentage profit.

Question 32

- The gradient of the line joining points $(-2, n)$ and $(n, 5)$ is $\frac{4}{3}$. Find the value of n .
- Given that $\tan x = \frac{15}{8}$, $0^\circ < x < 90^\circ$, find the value of $5\sin x - 6\cos x$.

Question 33

The graph shows the relation of the form $y = mx^2 + nx + r$, where m, n and r are constants. Using the graph:



- (a) State the scale used on **both** axes;
- (b) find the values of m, n and r .
- (c) find the gradient of the line through **P** and **Q**;
- (d) state the range of values of x for which $y > 0$.

Question 34

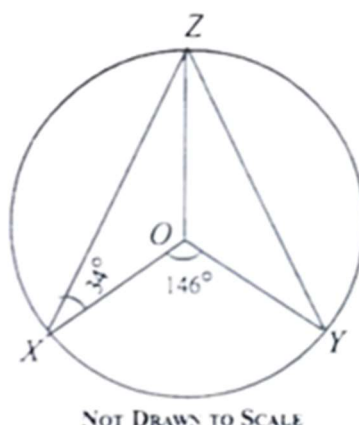
- (a) Copy and complete the table of values for $y = \sin x + 7 \cos x$ for $0^\circ \leq x \leq 180^\circ$.

x	0°	20°	40°	60°	80°	100°	120°	140°	160°	180°
y	7.0				4.2		-0.9			

- (b) Using a scale 2 cm to 20° on the x -axis and 2 cm to 2 units on the y -axis, draw the graph of $y = \sin x + 7 \cos x$ for $0^\circ \leq x \leq 180^\circ$.
- (c) Using the graph, find the:
- value of y when $x = 150^\circ$;
 - range of values of x for which $y > 0$.

Question 35

- (a) The exterior angles of a polygon are 42° , 38° , 57° , x° , $(x + y)^\circ$, $(2x - 15)^\circ$ and $(3x - y)^\circ$. If x is 7° less than y , find the values of x and y .
- (b)



In the diagram, O is the centre of the circle XYZ . $\angle ZOX = 34^\circ$ and $\angle XOY = 146^\circ$. Find $\angle OYZ$.

Question 36

- (a) The probability that an athlete will **not** win any of the **three** races is $\frac{1}{4}$.
If the athlete runs in all the races, what is the probability that the athlete will win:
- only** the **second** race;
 - all** the **three** races;
 - only two** of the races?
- (b) A cone with perpendicular height 24 cm has a volume of 1200 cm^3 . Find the volume of a cone with the same base radius and height 84 cm. [Take $\pi = \frac{22}{7}$]

Question 37

- (a) The diameter of a cylinder closed at both ends is 7 cm. If the total surface area is 209 cm^2 , calculate the height. [Take $\pi = \frac{22}{7}$]
- (b) The points X and Y, 19 m apart are on the same side of a tree. The angles of elevation of the top, T, of the tree from X and Y on the horizontal ground with the foot of the tree are 43° and 38° respectively.
- (i) Illustrate the information in a diagram.
- (ii) Find, correct to **one** decimal place, the height of the tree.

Question 38

An isosceles triangle PQR has its vertices on the circumference of a circle.

If $|PQ| = |QR| = 17 \text{ cm}$, $|PR| = 16 \text{ cm}$ and M is the midpoint of $|PR|$, calculate:

- (a) $|QM|$;
- (b) correct to the **nearest** whole number, the radius of the circle.

Question 39

An isosceles triangle PQR has its vertices on the circumference of a circle.

If $|PQ| = |QR| = 17 \text{ cm}$, $|PR| = 16 \text{ cm}$ and M is the midpoint of $|PR|$, calculate:

- (a) $|QM|$;
- (b) correct to the **nearest** whole number, the radius of the circle.

Question 40

- (a) Aman spent $\frac{1}{3}$ of his income on rent, $\frac{1}{6}$ of the income in a fast food shop and $\frac{3}{4}$ of the remaining amount was saved in a bank. If he had N125, 000.00 left, find :

- (i) his income;
- (ii) the amount saved in the bank.

- (b) Solve: $\frac{2+3x}{5} - \frac{1-7x}{3} < 2x + 9$.

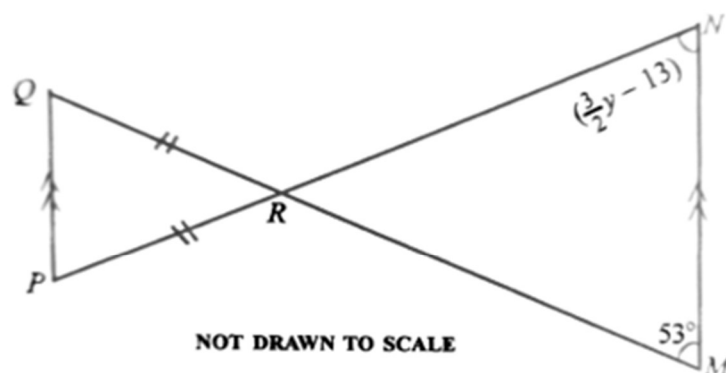
Question 41

- (a) A cylindrical tank of diameter 3.5 m is full of water. Three tankers, each of capacity 6,500 litres, draw water from the tank. Calculate, in metres, correct to **two** decimal places, the reduction in height of water in the tank. [Take $\pi = \frac{22}{7}$]

- (b) The coordinates of two points P and Q in a plane are (2, -5) and (-4, -6), respectively. Find the equation of the straight line joining the points.

Question 41

(a)



In the diagram, $\overline{PQ} \parallel \overline{MN}$, $\overline{PR} \parallel \overline{NR}$, $\angle QMN = 53^\circ$ and $\angle MNP = \left(\frac{3}{2}y - 13\right)^\circ$. Find:

- (i) $\angle PRQ$;
- (ii) the value of y .

(b) A bowl contains 15 yellow and green balls of the same size. If the probability of selecting a yellow ball is $\frac{2}{5}$, find the number of:

- (i) yellow balls;
- (ii) green balls in the bowl.

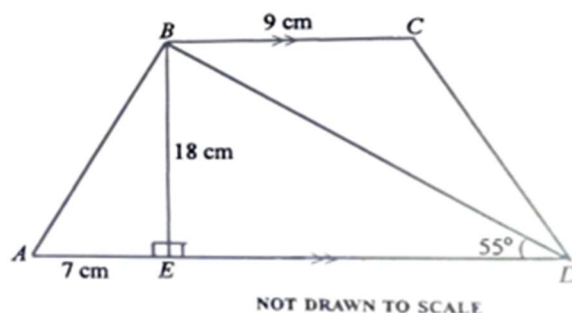
Question 42

Given that $S = \sqrt{\frac{5d(L-d)}{7}}$,

- (a) make L the subject of the relation;
- (b) find, correct to **two** decimal places, the value of L when $d = 27$ and $S = 0.7$.

Question 43

(a)



$ABCD$ is a trapezium with $\overline{BC} \parallel \overline{AD}$. E is a point on AD such that $\overline{BE}$ is perpendicular to $\overline{AD}$. If $\angle BDA = 55^\circ$, $|AE| = 7 \text{ cm}$, $|BE| = 18 \text{ cm}$ and $|BC| = 9 \text{ cm}$, find:

- $\angle BAE$;
- the area of $ABCD$.

(b) A trader bought goods for GH¢ x and sold them for GH¢ y . If the profit was 25%, find an equation connecting x and y .

Question 44

(a) A vertical pole 6 m high, cast a shadow 9 m long at the same time that a tree cast a shadow 30 m long. Find the height of the tree.

(b) Solve for y in the expression: $\log(5y - 4) = \log(y + 1) + \log 4$.

Question 45

The sides of a rectangle are $(x + 3) \text{ cm}$ and $(x - 2) \text{ cm}$. If the area of the rectangle is 36 cm^2 , find its:

- dimensions;
- perimeter.

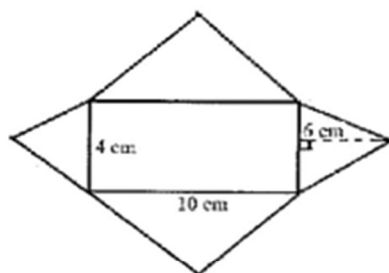
Question 46

The table shows the distribution of ages of some students in a school.

Age	12	13	15	10	14	16
Frequency	2	3	3	9	6	2

Find, correct to **one** decimal place, the mean deviation.

Question 47



NOT DRAWN TO SCALE

The diagram shows the net of a rectangular pyramid. Calculate, correct to **two** decimal places, the:

- (a) slant height;
- (b) perpendicular height;
- (c) total surface area of the pyramid.

Question 48

- (a) A three-digit number is such that the sum of the digits is 12 and the last digit is **three** Times the first one. If the middle digit is half the sum of the other two, find the number.
- (b) The 7th and 18th terms of an Arithmetic Progression (A.P.) are $15\frac{1}{4}$ and $42\frac{3}{4}$ respectively. Find the:
 - (i) n^{th} term;
 - (ii) sixth term,of the progression.

Question 49

The time (T), spent on erecting a concrete pillar is partly constant and partly varies as **three** times the base area (A). If it took 27 *hours* to erect a pillar with a base area of 24 m^2 and 52 *hours* for a pillar with a base area of 48 m^2 , find the time taken to erect a pillar with base area 60 m^2 .

Question 50

- (a) An aircraft **X** leaves airport **P** at 09:55 *hours* GMT and arrives at airport **Q** at 14:15 *hours* GMT. Given that the distance between the two airports is 2, 210 *km*, find the:
- flight time;
 - speed of the aircraft **X**.
- (b) Two years ago, a farmer purchased a machine for use in his farm. If the machine depreciates at an annual rate of $8\frac{1}{2}\%$ and the current value is ₦ 102, 500.00, find, correct to the **nearest** naira, the value of the machine:
- after the third year of the purchase;
 - when it was purchased.

Question 51

The table shows the marks scored by some candidates in an examination.

Mark (%)	1-10	11-20	21-30	31-40	41-50	51-60	61-70	71-80	81-90	91-100
Frequency	2	6	17	30	33	51	29	9	3	0

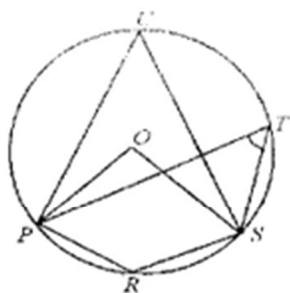
- Construct a cumulative frequency table.
- Draw a cumulative frequency curve.
- Using the graph, estimate the:
 - median;
 - pass mark if 60% of the candidates passed.

Question 52

- (a) The angle of depression of a boat from the midpoint of a vertical cliff is 35° . If the boat is 120 *m* away from the foot of the cliff, calculate, correct to **one** decimal place, the height of the cliff.
- (b) Using the quadratic formula, solve, correct to **three** significant figures, $2x^2 + 3x - 4 = 0$.

Question 53

(a)



NOT DRAWN TO SCALE

In the diagram, O is the centre of the circle $PRSTU$ and $\angle PTS = 58^\circ$. Find:

- (i) $\angle POS$;
- (ii) $\angle PRS$.

(b) Ato, Bonsu and Musah have a joint business venture. They agreed to raise the capital as follows: Ato GH¢ 2, 000.00 per month for 2 months; Bonsu GH¢ 3, 000.00 for one month and Musah GH¢ 2,500.00 per month for 3 months. It was also agreed that profit would be shared in proportion to the total amount contributed by **each** partner. If the profit at the end of a period was GH¢ 5, 220.00, calculate Ato's profit as a percentage of his contribution.

Question 54

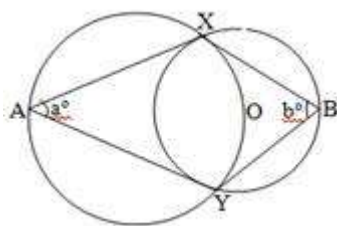
(a) There are 20 men on a public bus. Of these, 15 wear glasses and 10 wear wrist watches. If one man is chosen at random from the bus, what is the probability that he wears **both** glasses **and** wrist watches?

(b) *MNC* company buys a car for \$ 27, 000.00 and sells to Mr. Adams for \$ 36, 000.0 after a discount of 10% on the marked price. Calculate the:

- (i) marked price of the car;
- (ii) percentage profit made by the company.

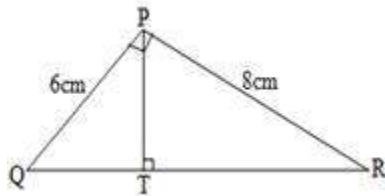
Question 55

(a)



In the diagram, the two circles intersect at X and Y. The centre of the smaller circle is on the circumference of the bigger circle. A and B are any two points on the major arcs, one on each circle. Find an equation connecting a and b.

(b)



In the diagram $\angle QPR = \angle PTR = 90^\circ$, $PR = 8\text{cm}$ and $PQ = 6\text{cm}$. Find TR .

Question 56

- (a) By how much is 110002 greater than or less than 1112×112 ?
- (b) A shopkeeper has in stock 20 televisions set percent. He sells 18 of them at a profit of 15 percent and the remaining two at a loss of 5 percentage. Find his profit on the 20 sets.

Question 57

The area of a rectangular football field is 7200m^2 while its perimeter is 360m . calculate the:

- . dimensions of the field;
- . cost of clearing the field at N6.50 per square meter, leaving a margin of 2m wide along the longer sides;
- . percentage of the part not cleared.

Question 58

In the diagram, a ladder LN 10m long, rests on a wall 4.5m high such that 2.5m of it projects beyond the wall.

Calculate, correct to one decimal place, the angle which the ladder makes with the ground.

How high above the ground is the upper end of the ladder?

If the foot of the ladder is moved by 2m further away from the wall, calculate, correct to the nearest degree, the angle which the ladder makes with the ground

Question 59

In a class of 200 students, 70 offered Physics, 90 Chemistry, 100 Mathematics while 24 did not offer any of the three subjects. Twenty three (23) students offered Physics and Chemistry, 41 Chemistry and Mathematics while 8 offered all three subjects.

Draw a Venn diagram to illustrate the information.

Find the probability that a student selected at random from the class offered:

Physics only;

Exactly two of the subjects.

Question 60

(a) Consider the statements:

p : Some teachers in a school are graduates.

q : Only graduate teachers received National honours.

(i) Represent this information in a Venn diagram.

(ii) If Mr. Sowah is a teacher in the school, determine whether or **not** the following conclusions are **valid** or **not valid**.

(I) Mr. Sowah is a graduate $\Rightarrow$ Mr. Sowah received National honours.

(II) Mr. Sowah received National honours $\Rightarrow$ Mr. Sowah is a graduate.

(III) Mr. Sowah did not receive a national honour $\Rightarrow$ Mr. Sowah is **not** a graduate.

(b) The n^{th} term of a sequence is 2^{2n+1} . Find the **first** 3 terms of the sequence.

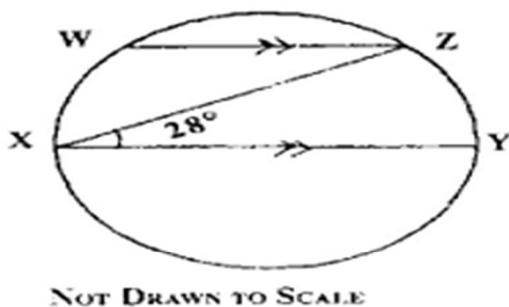
Question 61

(a) An aeroplane flies 500 km from town P on a bearing of 053° to town Q . It then flies 700 km to town R on a bearing of 165° .

(i) Illustrate the information with a diagram

(ii) Calculate, correct to **three** significant figures, the distance between P and R .

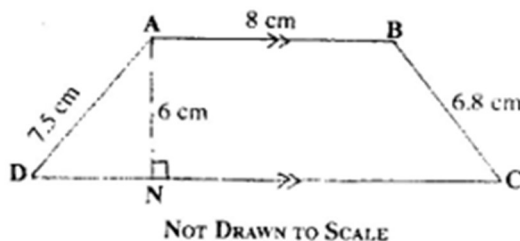
(b)



In the diagram, $\overline{XY}$ is the diameter of the circle $WXYZ$, $\overline{XY} \parallel \overline{WZ}$ and $\angle ZXY = 28^\circ$. Find $\angle XWZ$.

Question 62

(a)



The diagram, $ABCD$ is a trapezium in which $\overline{AB} \parallel \overline{DC}$, $|\overline{AD}| = 7.5 \text{ cm}$, $|\overline{AB}| = 8 \text{ cm}$, $|\overline{BC}| = 6.8 \text{ cm}$ and $|\overline{AN}| = 6 \text{ cm}$. Calculate the area of the trapezium.

- (b) The equation of the line passing through the point $(-3, -4)$ is $6y + kx - 12 = 0$. Find the value of k .

Question 63

- (a) Copy and complete the table of values for the relation $y = 2x^2 - 5x - 3$ for $-3 \leq x \leq 5$.

x	-3	-2	-1	0	1	2	3	4	5
y	30			-3				9	

- (b) Using a scale of 2 cm to 1 unit on the x axis and 2 cm to 5 units on the y axis, draw the graph of $y = 2x^2 - 5x - 3$ for $-3 \leq x \leq 5$.

- (c) Using the graph:

- find the truth set of $2x^2 - 4x - 7 = x - 4$
- determine the equation of the line of symmetry;
- find the values of x for which $y = 7$.

Question 64

- (a) In an isosceles triangle ABC , $|\overline{AB}| = |\overline{AC}| = 6 \text{ cm}$ and $\angle BAC = 130^\circ$. Find, correct to **two** significant figures, the:

- $|\overline{BC}|$;
- area of the triangle ABC .

- (b) On a circular park with diameter of 50.0 m, there are 10 lamps whose bases are circles with radius 0.5 m. The entire area of the park is covered with grass except the bases of the lamps. Calculate the area of the park covered by the grass.

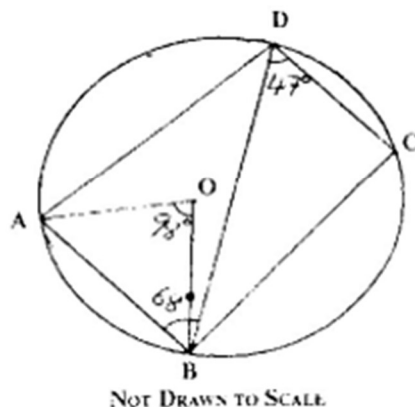
[Take $\pi = \frac{22}{7}$]

Question 65

- (a) An aeroplane flies 100 km from point A to point B on a bearing of 330° . It then flies from point B to point C 300 km due west.
- Illustrate this on a diagram.
 - How far west, correct to the nearest km, is the aeroplane from the starting point?
- (b) A student added consecutive odd numbers starting from 11 and had a sum of 551. How many odd numbers were added?

Question 66

(a)



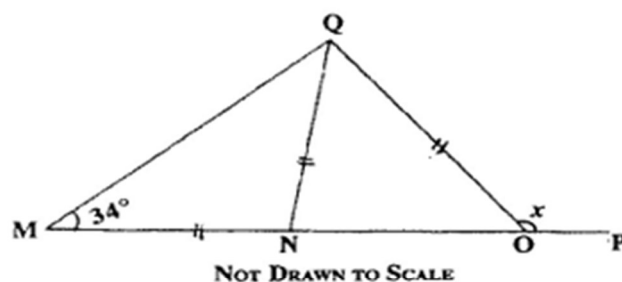
NOT DRAWN TO SCALE

In the diagram, O is the centre of circle ABCD such that $\angle AOB = 98^\circ$, $\angle DBA = 68^\circ$ and $\angle BDC = 47^\circ$. Find:

- $\angle CBD$;
 - $\angle DCB$.
- (a) Given that $10\cos(x+17) - 2 = 0$, $0^\circ \leq x \leq 90^\circ$, calculate, correct to the nearest degree, the value of x .
- (b) The sum of the interior angles of a regular polygon with n sides is $(120n)^\circ$. Find the value of n .

Question 67

(a)



In the diagram, $QMN = 34^\circ$, $|MN| = |NQ| = |QO|$ and $\angle QOP = x$. Find the value of x .

(a) A box contains 3 red balls and 4 white balls. Two balls are picked at random one after the other **without replacement**. Find the probability that the two balls picked are:

- (i) **both** red;
- (ii) **both** white;
- (iii) of the **same** colour;
- (iv) of **different** colours.

Question 68

A triangular plot of land ABC is such that $|AB| = 85\text{ m}$, $|BC| = 110\text{ m}$ and $\angle CAB = 60^\circ$,

- (a) Using a ruler and a pair of compasses **only** and a scale of 1 cm to 10 m , construct the:
- (i) triangular plot ABC ;
 - (ii) position of a vertical telegraph post X which is equidistant from AC and BC and perpendicular from B to AC .

(b) Find the actual distance from X to C .

Question 69

- (a) Two fractions have the same denominator, 8. The sum of the two fractions is $\frac{1}{2}$. If one of the fractions is added to 5 times the other, the result is 2. Find the two fractions.
- (b) If $a = \frac{mg - kv^2}{m}$, find, correct to the nearest whole number the value of v when $a = 2.8$, $m = 12$, $g = 9.8$ and $k = \frac{8}{3}$.

Question 70

- (a) Given that $\cos 60^\circ = \sin 30^\circ = \frac{1}{2}$ and $\cos 30^\circ = \sin 60^\circ = \frac{\sqrt{3}}{2}$, evaluate $\frac{\tan 60^\circ - 1}{1 - \tan 30^\circ}$
- (b) In a class of 40 students, 30 read Chemistry and 20 read Physics. If all the students read at least one of the subjects, find the probability that a student selected at random from the class reads **only** Chemistry.

Question 72

A ladder 10 m long, leans against a vertical wall at an angle of 70° to the ground. If the ladder slips down the wall 4 m, find, correct to two significant figures:

- (a) the new angle which the ladder makes with the ground;
- (b) the distance the ladder slipped back on the ground from its original position.

Question 73

Question 74

Question 75

Question 76

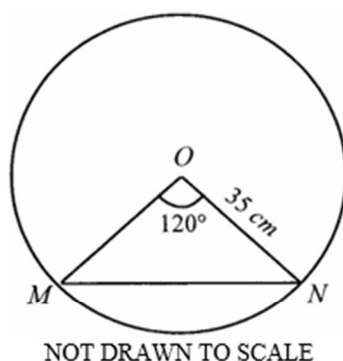
Question 77

- (a) Differentiate $y = 3t^4 - 4t^3 + 2t^2 - \frac{1}{t} - 5$
- (b) The probability that a civil servant owns a car is $\frac{1}{6}$. If two civil servants are selected at random, find the probability that: (i) **each** owns a car; (ii) **only** one owns a car.
- (c) Solve: $\log_{10}(16x + 2) - \log_{10}(4x + 2) = \log_{10}(2x + 4)$.

Question 78

- (a) The ratio of the radius(r), of the base of a cone to the height (h) is 2:3. If the height of the cone is 8.1 cm, calculate, correct to one decimal place, the volume of the cone.[Take $\pi = \frac{22}{7}$]
- (b) Mr. Uduh invested an amount of money in two separate finance firms in the ratio 3:5. His profit was calculated on 8% and 10% rate of simple interest respectively. If after a year, he received a sum of N92, 000.00 as total interest, calculate, correct to the **nearest** whole number, the total amount invested.

Question 79



The diagram is a circle, centre O with radius 35 cm. The arc MN subtends an angle of 120 at the centre. Find, correct to **one** decimal place, the:

- perimeter of the minor sector MON;
- length of the chord MN;
- area of the minor segment cut off by the chord MN. [Take $\pi = \frac{22}{7}$]

Question 80

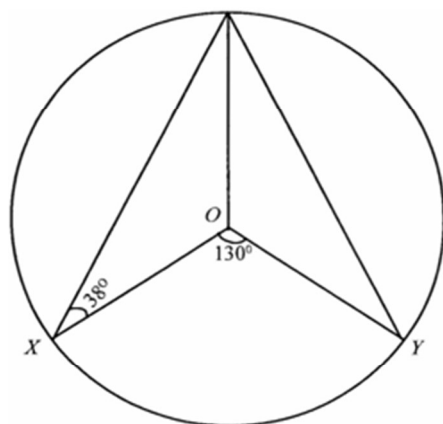
- Given that $\cos(2y - 16) = \frac{1}{2}$, $0 \leq y \leq 90$, find the value of y .
- Using a ruler and a pair of compass only,
 - Construct a triangle ABC in which $|AB| = 8$ cm, $|BC| = 9$ cm and $\angle ABC = 75^\circ$.
 - Locate the point P inside ABC such that $|PA| = |PB|$ and $|PA| = 4.5$ cm.
 - Measure:
 - $|CP|$;
 - $\angle ACB$.

Question 81

(a) The area of a triangle is 54 cm^2 . If the height is 3 cm more than the base, find the length of the base.

Z

(b)



NOT DRAWN TO SCALE

In the diagram, O is the centre of circle. The points X, Y and Z are on the circumference of the circle, $\angle ZXO = 38^\circ$ and $\angle XOY = 130^\circ$. Find:

(i) $\angle OYZ$;

(ii) $\angle ZOY$.